

Tactical DNA: Coaches, Styles, and Player Roles from Football Passing Networks

EECS 4414 Project Proposal

Mahyar Jaberi
mhryrjbr@my.yorku.ca
York University
Toronto, Ontario, Canada

Keywords

network analysis, football analytics, passing networks, social network analysis, motif analysis, sports analytics

1 Motivation and Problem Definition

Football tactics are extensively debated in informal media but rarely quantified with rigor. Coaches, pundits, and fans speak of “possession football,” “gegenpressing,” or a team’s “style of play” as if these were measurable properties, yet such labels are typically applied qualitatively. Passing networks, defined as directed weighted graphs whose nodes represent players and whose edge weights count completed passes from one player to another, provide a principled way to encode tactical structure as a measurable mathematical object [2, 7]. A growing body of work has shown that passing network properties relate to team performance and offensive efficiency [1, 4], but most studies are descriptive: they characterize a single team or compare a small number of matches, without using network signatures as features for higher-level inference tasks.

We propose to close this gap by treating passing networks as *tactical fingerprints* and using them to answer three concrete inferential questions. Specifically, we draw on StatsBomb’s open event data [8] covering the La Liga “Big Three Golden Era” from 2008/09 to 2015/16, focusing on FC Barcelona, Real Madrid, and Atlético Madrid, supplemented by the 2015/16 Bundesliga season at Bayern Munich and the men’s FIFA World Cups of 2018 and 2022. This yields approximately 800 team-match passing networks across roughly ten identifiable head coaches and three competitive contexts. Formally, let $G_m = (V_m, E_m, w_m)$ denote the passing network of a team in match m , where V_m is the set of starting players, $E_m \subseteq V_m \times V_m$ is the set of directed pass relations, and $w_m : E_m \rightarrow \mathbb{Z}^+$ assigns each edge the number of completed passes. Let $\phi(G_m) \in \mathbb{R}^d$ be a feature embedding of G_m derived from its structural properties (centrality distributions, motif counts, density, and clustering coefficient). We address:

Problem 1 (Coach Identification). Given a labeled set $\{(G_m, c_m)\}$ where c_m is the coach of the team in match m , learn a classifier $f : \mathbb{R}^d \rightarrow \mathcal{C}$ such that $f(\phi(G_m))$ predicts c_m with accuracy significantly above the team-identity baseline. We are particularly interested in whether the signature of a coach persists when the same coach moves between clubs, most notably evaluating whether Pep Guardiola’s tactical fingerprint is preserved across his Barcelona tenure (2008/09 to 2011/12) and his 2015/16 season at Bayern Munich. This transfer-learning question directly tests whether the coach signal in passing networks is club-bound or coach-bound.

Problem 2 (Tactical Archetype Discovery). Given an unlabeled set $\{G_m\}$, partition the matches into clusters $\{C_1, C_2, \dots, C_k\}$

that optimize an internal cluster-quality measure (silhouette, modularity). We then evaluate whether the discovered clusters correspond *a posteriori* to known tactical schools such as possession-based play, gegenpressing, or low-block counter-attack.

Problem 3 (Player Role Embedding). For each player p , construct an aggregated ego-network signature $\psi(p) \in \mathbb{R}^{d'}$ summarizing the player’s local network role across all matches in which they appeared. Learn a low-dimensional embedding $\Phi : \mathbb{R}^{d'} \rightarrow \mathbb{R}^2$ that preserves structural similarity, evaluated by *position purity* — the fraction of each player’s k nearest neighbors that share their on-pitch position. We complement this quantitative measure with qualitative case studies, identifying structural successors to canonical playmakers such as Xavi Hernández, Andrea Pirlo, and Luka Modrić.

2 Methodology

2.1 Data and Network Construction

We use the publicly available StatsBomb Open Data [8], which provides professional-grade event-level records for several major competitions. Our analysis draws on three data sources: the La Liga seasons from 2008/09 to 2015/16 (covering FC Barcelona, Real Madrid, and Atlético Madrid across the “Big Three Golden Era”), the 2015/16 Bundesliga season (covering Pep Guardiola’s final year at Bayern Munich), and the men’s FIFA World Cups of 2018 and 2022. Each event is annotated with a timestamp, location on a standardized 120×80 pitch, the players involved, and outcome.

For each match m and each team, we construct the passing network G_m by aggregating all completed open-play passes during the period in which the eleven starting players are on the pitch. Following established practice [2, 7], we restrict nodes to the eleven starters to ensure comparable graph sizes across matches and to avoid the structural distortion introduced by substitutions. Each directed edge (u, v) is weighted by the number of completed passes from player u to player v .

2.2 Network Features

The feature embedding $\phi(G_m)$ aggregates three families of structural descriptors, each shown in prior work to carry tactical signal:

Centrality measures. For each player node $v \in V_m$, we compute weighted in-degree and out-degree, betweenness centrality, eigenvector centrality, and PageRank. From the resulting per-node distributions we extract summary statistics (mean, variance, skewness, Gini coefficient), yielding a centralization profile that captures how hierarchically organized the team’s passing is [2].

Global structural metrics. Network density, weighted clustering coefficient, average shortest-path length, algebraic connectivity

(the second-smallest eigenvalue of the graph Laplacian), and the largest eigenvalue of the adjacency matrix. These collectively capture how cohesive a team’s passing organization is.

Network motifs. Motifs are small recurring subgraph patterns whose frequencies characterize the local structural “vocabulary” of a graph [6]. We enumerate all 13 connected directed 3-node motifs and the most common 4-node motifs, recording their normalized frequencies. Recent work has demonstrated that motif distributions differentiate tactical play styles — for example, possession-based teams produce disproportionately many 3-cycles (the “rondo” pattern), while counter-attacking teams produce more linear chain motifs [4].

For Problem 3, we construct a separate per-player feature vector $\psi(p) \in \mathbb{R}^{d'}$ from each player’s ego-network properties aggregated across all matches in which they appeared. Ego-network features include ego degree, ego density, ego clustering coefficient, in/out weighted degree, motif-position frequencies (e.g., how often the player occupies the middle of a chain motif), and the spatial dispersion of pass endpoints.

2.3 Algorithms

For **Problem 1**, we train three supervised classifiers using $\phi(G_m)$ as input and coach identity as the target: logistic regression as an interpretable linear baseline, a random forest to capture non-linear feature interactions, and XGBoost as a state-of-the-art gradient-boosted ensemble. All models are evaluated using stratified 5-fold cross-validation to provide a robust estimate of generalization performance and to handle class imbalance arising from unequal match counts per coach.

For **Problem 2**, we apply k -means clustering and hierarchical agglomerative clustering on $\phi(G_m)$, selecting k using both the silhouette score and the elbow method to triangulate the natural number of tactical archetypes. We visualize the resulting clustering using UMAP [5] projections of the feature space to enable qualitative interpretation against known tactical schools.

For **Problem 3**, we project the per-player ego-network features $\psi(p)$ into a low-dimensional role space using two complementary dimensionality-reduction techniques: PCA for an interpretable linear projection and UMAP for a non-linear projection that better preserves local neighborhood structure. k -nearest-neighbor retrieval in the resulting role space identifies players with structurally similar tactical roles.

All analysis is implemented in Python, with NetworkX for graph construction and feature computation, scikit-learn for classical machine learning models and clustering, XGBoost for gradient boosting, and umap-learn for the UMAP projections. The full codebase, along with reproducible scripts and processed data artifacts, will be released as a public GitHub repository.

3 Evaluation

3.1 Datasets

The evaluation corpus comprises all team-match passing networks constructed from our three data sources: the La Liga seasons from 2008/09 to 2015/16 (FC Barcelona, Real Madrid, and Atlético Madrid across the “Big Three Golden Era”), the 2015/16 Bundesliga season (Bayern Munich under Pep Guardiola), and the men’s FIFA World Cups of 2018 and 2022. This yields approximately 800 team-match

networks across roughly ten identifiable head coaches, including Pep Guardiola, Tito Vilanova, Tata Martino, and Luis Enrique at Barcelona; Manuel Pellegrini, José Mourinho, Carlo Ancelotti, and Rafa Benítez at Real Madrid; Diego Simeone at Atlético Madrid; and Pep Guardiola at Bayern Munich. Dozens of additional national-team coaches in the World Cup data provide tactical diversity for the unsupervised clustering analysis in Problem 2.

3.2 Evaluation Plan

Problem 1. We report classification accuracy and per-class macro-F1 with stratified 5-fold cross-validation, and we publish a confusion matrix to expose which coaches’ signatures are confused with which. The principal baseline is a team-identity classifier trained on the same features: if coach identity is a stronger signal than team identity, our coach classifier should outperform it. A second diagnostic experiment trains the model on Pep Guardiola’s Barcelona matches and tests on his Bayern Munich season (or vice versa), directly measuring whether the coach signature transfers across clubs and leagues. A third experiment ablates feature families (centrality, motifs, combined) to identify the most discriminative descriptors.

Problem 2. We report silhouette score, modularity, and average intra-cluster versus inter-cluster Euclidean distance across candidate values of k . We then manually inspect each cluster: examining the coaches, leagues, and tactical contexts of the matches it contains, and testing whether human-assigned tactical categories (possession-based, gegenpressing, low-block counter, direct) agree with the discovered partition. A UMAP projection of the feature space with matches annotated by landmark games from our corpus — including marquee Clásicos from the 2010–2013 Pep-vs-Mourinho era and the 2022 FIFA World Cup Final — serves as the principal qualitative figure.

Problem 3. For quantitative validation we test whether nearest-neighbor retrieval in the learned role space respects player position: a midfielder’s k nearest neighbors should be predominantly midfielders. We report position purity at $k = 5$ and $k = 10$, comparing against a random-placement baseline. We complement this with qualitative case studies: identifying structural successors to canonical playmakers (Xavi Hernández, Andrea Pirlo, Luka Modrić) by inspecting their nearest neighbors in the role space and discussing whether the discovered analogues align with widely-held tactical interpretations.

3.3 Roadmap, Risks, and Trade-offs

We identify three principal risks. First, class imbalance in coach labels is significant: Pep Guardiola accounts for far more matches than coaches with single-season tenures (e.g., Tito Vilanova, Tata Martino). We mitigate this through class-weighted training objectives and stratified sampling during cross-validation. Second, motif counts are sensitive to network size; we control for this by restricting our analysis to the eleven-starter sub-period of each match, ensuring all passing networks have exactly eleven nodes. Third, UMAP projections preserve local structure better than PCA but are harder to interpret semantically; we therefore report both. Time permitting, we will also explore a temporal extension — constructing per-segment passing networks within each match (e.g., 15-minute windows) to capture within-game tactical adaptation [3].

References

- [1] Ricardo Alves, Gonalo Dias, Nuno Andr  Nunes, S rgio M. Querido, and Vasco Vaz. 2025. Social Network Analysis in Football: A Systematic Review of Performance and Tactical Applications. *Frontiers in Psychology* 16 (2025), 1659603. doi:10.3389/fpsyg.2025.1659603
- [2] Javier M. Buld , Javier Busquets, Johann H. Mart nez, Jos  L. Herrera-Diestra, Ignacio Echegoyen, Javier Galeano, and Jordi Luque. 2018. Using Network Science to Analyse Football Passing Networks: Dynamics, Space, Time, and the Multilayer Nature of the Game. *Frontiers in Psychology* 9 (2018), 1900. doi:10.3389/fpsyg.2018.01900
- [3] Andr s Chacoma. 2025. Identification and Optimization of High-Performance Passing Networks in Football. *Physical Review E* 111 (2025), 044313. doi:10.1103/PhysRevE.111.044313
- [4] Ming-Xia Li, Li-Gong Xu, and Wei-Xing Zhou. 2024. Motif Analysis and Passing Behavior in Football Passing Networks. *arXiv preprint arXiv:2408.07927* (2024). <https://arxiv.org/abs/2408.07927>
- [5] Leland McInnes, John Healy, and James Melville. 2018. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. *arXiv preprint arXiv:1802.03426* (2018). <https://arxiv.org/abs/1802.03426>
- [6] Ron Milo, Shai Shen-Orr, Shalev Itzkovitz, Nadav Kashtan, Dmitri Chklovskii, and Uri Alon. 2002. Network Motifs: Simple Building Blocks of Complex Networks. *Science* 298, 5594 (2002), 824–827. doi:10.1126/science.298.5594.824
- [7] Javier L pez Pe a and Hugo Touchette. 2012. A Network Theory Analysis of Football Strategies. *arXiv preprint arXiv:1206.6904* (2012). <https://arxiv.org/abs/1206.6904>
- [8] StatsBomb. 2024. StatsBomb Open Data. GitHub repository. <https://github.com/statsbomb/open-data> Accessed: 2026-05-11.